

SARDAR PATEL UNIVERSITY
External Examination (CBCS)
M. Sc. - Information Technology (Integrated)
IVth Semester (CBCS)
PS04CIIT02 - Data Communication & Networking
17th April, Monday - 2017

Time : 10:00 am to 1:00 pm

Total Marks :70

Q-1 Select an appropriate option.

10

1. "Interconnected collection of autonomous computers" is known as _____.
(a) Computer Network (b) Host (c) Subnet (d) Server
2. Number of data elements sent from one device to another per second is known as _____.
(a) Bandwidth (b) Modulation Rate (c) Data Rate (d) None of these
3. A computer network permits sharing of _____.
(a) Resources (b) Information (c) Both (a) & (b) (d) None of these
4. Which of the following modulation requires the lowest bandwidth?
(a) ASK (b) PSK (c) FSK (d) QPSK
5. Transmission media are usually categorized as _____.
(a) Fixed or Unfixed (b) Guided or Unguided
(c) Determinate or Indeterminate (d) Metallic or Nonmetallic
6. The _____ layer lies between the network layer and the application layer.
(a) Data link (b) Physical (c) Transport (d) None of these
7. In Carrier Sense Multiple Access , if the station senses the medium before trying to use it then the chance of collision can be _____.
(a) Increased (b) Reduced (c) Highlighted (d) Decreased
8. For Carrier Sense Multiple Access/Collision Detection, we need a restriction on the _____.
(a) Collision Size (b) Signal Size (c) Frame Size (d) Station Size
9. The path that a satellite makes around the world is called _____.
(a) An orbit (b) A period (c) A footprint (d) An uplink
10. _____ is an example of circuit switching .
(a) Wireless (b) Telephone network
(c) Computer-to-Computer (d) Terminal-to-Computer

Q-2 Answer the following questions. (Attempt any TEN) 20

1. Define Computer Networks.
2. Differentiate between Broadcasting and Multicasting.
3. Differentiate between Bridge and Repeater.
4. List the Modulation Techniques.
5. List the seven layers of OSI.
6. Write the full form: (i) DNS (ii) SMTP
7. What is IP Datagram?
8. Draw Figure for TCP Header.
9. Write full form of CSMA/CD and CSMA/CA.
10. Write the full form (i) FTP (ii) HTTP
11. What is satellite? List categories of satellite.
12. What is Physical topology and Logical topology?

Q-3

- (a) Discuss the importance of computer networking. 5
- (b) What is LAN? List the characteristics of LAN. 5

OR

Q-3

- (a) Write short note on fiber optics transmission media. 5
- (b) Explain star topology with its advantages and disadvantages. 5

Q-4

- (a) What is switching? Explain Packet switching in detail. 5
- (b) What do you mean by Multiplexing? Differentiate between TDM and FDM. 5

OR

Q-4

- (a) What is switching? Explain Circuit switching in detail. 5
- (b) Differentiate between OSI and TCP/IP. 5

Q-5

- (a) Write a short note on TCP Segment Header. 5
- (b) Write a short note on CSMA/CD. 5

OR

Q-5

- (a) Write a short note on IPv4 protocol. 5
- (b) Explain TCP Connection Establishment. 5

Q-6 Write short note: (i) Wireless Networks (ii) Microwave Transmission 10

OR

Q-6 Write short note: (i) Geosynchronous satellite (ii) Radio Transmission 10

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